


$$\text{Profits} = P \cdot Q - \underset{\substack{\uparrow \\ \text{AVC}}}{C(Q)} \cdot Q = \underbrace{(P - C(Q))} \cdot Q$$

$$\text{Profits} \geq 0 \quad \text{if and only if} \quad P \geq C(Q)$$

$$Pr(P \geq C(Q))$$

$$f_{\omega}(y) = \sim I_{[0, \omega)}(y)$$

$$E(Y) = \int_{-\infty}^{\infty} y \cdot (\sim) I_{[0, \omega)}(y) dy$$

$$= \int_0^{\infty} y \cdot (\sim) dy$$

PDF's : $\int_{-\infty}^{\infty} f_x(x) dx = 1$

$$1 = \int_{-\infty}^{\infty} \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} I_{[0, \infty)}(x) dx$$

$$\Gamma(\alpha) = \int_0^{\infty} \beta^\alpha x^{\alpha-1} e^{-\beta x} dx$$

↙ If α is a natural number

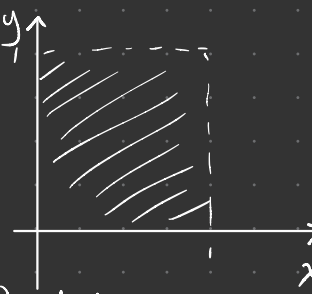
$$\Gamma(\alpha) = (\alpha-1)!$$

$$E(X) = \int_{-\infty}^{\infty} x \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} I_{[0, \infty)}(x) dx$$

$$= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} \int_0^{\infty} \underbrace{\beta^\alpha x^{(\alpha+1)-1} e^{-\beta x}}_{\Gamma(\alpha+1)} dx$$

PDF of a
Gamma dist. with
parameters $(\alpha+1, \beta)$

$$= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} = \frac{\alpha!}{(\alpha-1)!} = \frac{\alpha(\alpha-1)!}{(\alpha-1)!} = \alpha$$

$$f(x, y) = (x+y) I_{[0,1]}(x) I_{[0,1]}(y)$$


$$\begin{aligned} E[X^m Y^n] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^m y^n \cdot (x+y) I_{[0,1]}(x) I_{[0,1]}(y) dx dy \\ &= \int_0^1 \int_0^1 (\underline{x^{m+1} y^n} + x^m y^{n+1}) dx dy \end{aligned}$$

$$\begin{aligned} \int_0^1 \int_0^1 x^{m+1} y^n dx dy &= \int_0^1 y^n \left(\int_0^1 x^{m+1} dx \right) dy \\ &= \int_0^1 y^n \left[\frac{x^{m+2}}{m+2} \Big|_0^1 \right] dy = \int_0^1 y^n \left(\frac{1}{m+2} \right) dy \\ &= \frac{1}{m+2} \cdot \frac{y^{n+1}}{n+1} \Big|_0^1 = \frac{1}{m+2} \cdot \frac{1}{n+1} \end{aligned}$$

Law of the unconscious statistician

X , $g(X)$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$